

GOVERNANCE DESIGN WHITE PAPER

Laminar Democracy

A design and simulation study of tiered direct democracy with adaptive epistemic friction

Voting friction is matched to the epistemic demand of each question: a universal basic franchise for civic matters, certified electorates for technical ones, and status-priced agenda access, evaluated by a paired agent-based simulation.

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Simulation code and data

github.com/snowflake-amiya/laminar-democracy

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Abstract

Direct democracy applies a single, uniform decision procedure to questions of wildly varying epistemic difficulty, and the costs of that uniformity fall exactly where the stakes are highest: on technical questions where the median voter's individual accuracy is near chance and errors are correlated by shared misinformation. This paper develops **tiered direct democracy with adaptive epistemic friction** (TDD), an institutional design in which the franchise on highly technical questions is conditional on demonstrated domain competence, while a universal basic franchise is retained for ordinary civic questions. Status is acquired through open, criterion-referenced, domain-scoped certification with mandatory decay, and referendum signature thresholds fall with the initiator's status, creating a price signal for civic investment. We evaluate the design with a paired agent-based simulation ($N = 2,000$; 240-300 replications) comparing pure direct democracy, a two-tier variant, a three-tier variant, and a briefed citizens' assembly. The three-tier design attains 0.992 decision accuracy on highly technical issues against 0.536 under direct democracy, dominates the two-tier design at every certification validity tested, and is most advantageous precisely when certification tests are noisy. Capture experiments show certified tiers are nearly immune to mass-persuasion attacks but expose a bribery surface that a democratic veto overlay largely neutralizes. We conclude that the three-tier design is preferred under the paper's pre-registered decision rule, conditional on test-governance institutions that the design itself specifies but whose political feasibility remains an open question.

1 Introduction

Democratic systems routinely decide questions of radically different epistemic character. On the same ballot, citizens may be asked to judge a constitutional value question, a municipal budget, and the safety threshold of a reactor coolant system. Existing democratic architectures, whether representative or direct, apply an essentially uniform decision procedure to all of them: every eligible citizen casts one equally weighted vote, regardless of the question's technical content. This uniformity has a well-documented epistemic cost. Because the probability of any one vote being pivotal is vanishingly small, citizens have little incentive to invest in the specialized knowledge that technical questions demand (Downs, 1957), and the resulting deficits are systematic rather than random (Caplan, 2007). Direct democracy, which removes the filtering layer of representation, exposes this problem in its purest form.

This paper proposes and evaluates an alternative: **tiered direct democracy with adaptive epistemic friction**. The design rests on a simple principle: the friction a citizen must overcome to participate in a decision should scale with the epistemic demand of that decision, rather than remaining uniform across all decisions. Operationally, the design separates the agenda into three domains. Domain A questions (basic civic matters such as rights, values, and ordinary legislation) are decided by the entire citizenry. Domain B questions (intermediate technical matters such as budgets and infrastructure) are decided by citizens holding tier-2 status, acquired through an open certification test. Domain C questions (highly technical matters) are decided by citizens holding tier-3 status, acquired through a more demanding certification. In parallel, the initiative process is itself tiered: any citizen may propose a basic referendum by collecting a fixed number of

signatures, while status-holders may propose tiered referendums at lower signature thresholds drawn from their own tier. Status is therefore an earned and revocable license to operate on harder questions, not a general elevation of civic standing.

The paper makes three contributions. First, it formalizes the institutional design, extending the core two-tier proposal with domain-scoped certification, mandatory status decay, criterion-referenced (rather than quota-based) cutoffs, and a democratic veto overlay, each of which responds to a specific objection in the political-theory literature. Second, it reports a simulation study that compares the two-tier and three-tier variants under a pre-registered decision rule, alongside benchmarks of pure direct democracy and a briefed citizens' assembly. Third, it situates the design in its intellectual lineage and analyzes the strongest objections against it, particularly those concerning disenfranchisement, political equality, and the classification power implied by any domain boundary.

The remainder of the paper proceeds as follows. Section 2 characterizes the problem of uniform epistemic friction. Section 3 reviews precedents from Athens to deliberative assemblies. Section 4 specifies the institution. Section 5 presents the simulation model and results. Sections 6 and 7 analyze objections and compare alternatives, and Sections 8 and 9 discuss limitations and conclude.

2 The Problem of Uniform Epistemic Friction

The rational-ignorance argument is now half a century old and remains the sharpest statement of the problem. A voter's expected return on investing in technical policy knowledge is effectively zero, because the probability that one vote changes the outcome is negligible (Downs, 1957). The deficit is therefore not a failure of character but a rational response to incentives, and it deepens as questions become more technical. Caplan (2007) shows the deficit is systematic: voters hold biased beliefs about economics that do not wash out in aggregation, because the same biases are shared across voters. Achen and Bartels (2016) extend the point to identity and partisanship, arguing that retrospective voting rewards and punishes officials for events they did not control. Together these results undermine a comfortable picture in which individually noisy but independent judgments aggregate into collective wisdom.

That comfortable picture is the Condorcet jury theorem and its descendants. If each voter is more likely than not to be correct (competence above one half) and errors are independent across voters, then majority rule approaches certainty as the electorate grows (Condorcet, 1785; Surowiecki, 2004; Landmore, 2013). Both conditions fail exactly where stakes are highest. On highly technical questions the median citizen's accuracy is plausibly at or below one half, and in modern information environments errors are correlated, driven by shared media frames and active misinformation campaigns. Correlated error is fatal to the theorem's asymptotic guarantee: a population whose members all consult the same wrong source votes wrong together, no matter how large it is. Landmore (2013) adds that cognitive diversity, not raw ability, is the engine of collective intelligence, which complicates any naive remedy of filtering for the ablest subset.

Representative systems respond to this problem by delegation: independent central banks, regulatory agencies, and judicial review insulate technical decisions from mass preference, at a well-known cost in democratic legitimacy and accountability. The result is a standing dilemma. Competence is purchased by removing decisions from popular control; legitimacy is purchased by returning them to it. We use the term **epistemic friction** for the aggregate procedural cost a citizen must bear to participate competently in a decision: the cost of registration and turnout, of gathering signatures, of deliberation, and above all of acquiring the domain knowledge the question demands. Existing systems set this friction uniformly. The design explored here makes it adaptive. One further problem deserves immediate notice: if friction is to vary by domain, someone must classify questions into domains, and that classification is itself an object of political struggle. Section 6 treats this boundary problem directly; it is the design's most serious exposure.

3 Lineage: Precedents and Theoretical Neighbors

Competence-screened participation is older than modern democracy. Classical Athens combined an unusually broad citizen franchise with dense epistemic filtering for specific functions: most magistracies were filled by lot, but nominees underwent *dokimasia*, a formal scrutiny of competence and character before assuming office, and technical offices such as military commands were filled by election (Hansen, 1991). Ober (2008) reads classical Athens as a knowledge-producing machine whose institutions aggregated dispersed expertise while keeping authority broadly distributed. The Athenian precedent is instructive for the present design in one specific respect: competence checks operated on particular tasks, never as a general grade of citizenship, and the assembly itself remained open to all citizens.

The Venetian Republic's procedure for electing the doge is the canonical historical example of multi-stage epistemic filtering. Its intricate protocol alternated random selection with progressively smaller qualified colleges precisely to defeat factional capture, and it remained stable for five centuries. Modern sociocracy embeds a weaker version of the same idea through consent-based, domain-scoped decision rounds. The Swiss initiative system, the closest living analogue to the referendum component proposed here, prices agenda access through signature thresholds (100,000 signatures within 18 months for a constitutional initiative), but prices it uniformly: a Nobel laureate and a first-time petitioner face identical thresholds. The design in Section 4 makes that price status-graded.

The deliberative wave contributes the most direct contemporary evidence that competence is manipulable by institutional design. Fishkin's deliberative polls show large, systematic opinion changes after citizens are briefed on technical issues, suggesting that the raw survey electorate is not measuring an incorrigible ignorance but an unmobilized capacity (Fishkin, 2009). Citizens' assemblies recruited by sortition have since produced consequential policy on abortion, marriage equality, and climate in Ireland, Iceland, France, and elsewhere, and now operate at national scale in several jurisdictions (OECD, 2020). Landmore (2020) generalizes these experiments into a proposal for lottocratic chambers standing alongside elected ones. Liquid democracy offers a complementary mechanism: citizens may delegate their vote issue by issue and revoke delegation at will, effectively letting each citizen choose their own epistemic friction, though empirical deployments show delegation concentrating into a small number of super-delegates (Blum &

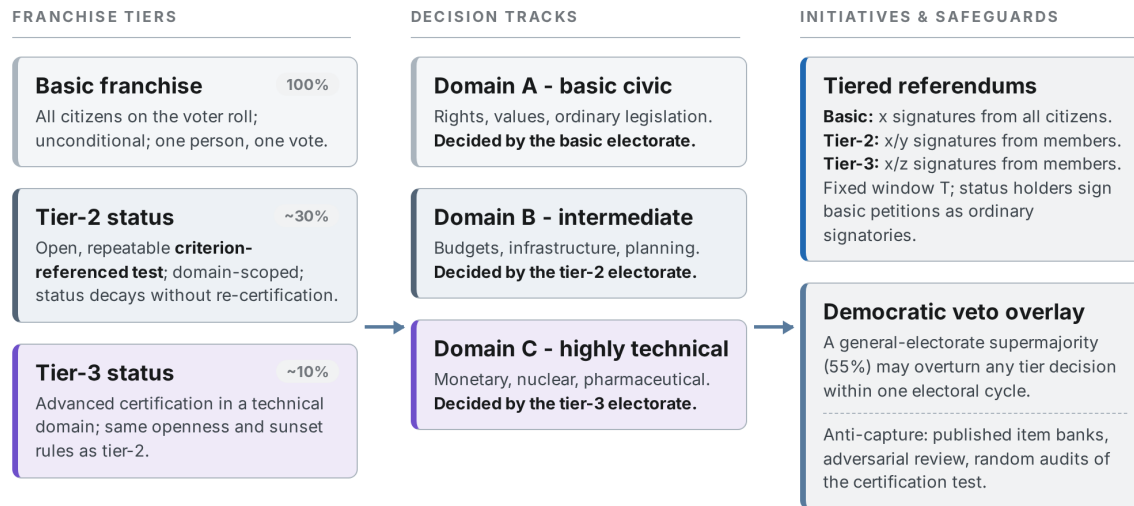
Zuber, 2016).

The epistocratic literature supplies the design's most direct theoretical rival. Its earliest modern form is Mill's (1861) case for universal but graduated suffrage, which would attach plural votes to demonstrated education rather than to property, with public examination as the preferred test. Brennan (2016) argues from a competence principle that the franchise itself should track knowledge, for example through plural weighting by tested competence. The objections to epistocracy are equally well developed: the history of literacy tests and grandfather clauses in the United States demonstrates that competence tests, once deployed as franchise gates, were engineered to disenfranchise Black voters and immigrants (Keyssar, 2000), and Hong and Page (2004) prove that under broad conditions a randomly selected group of diverse problem solvers outperforms a group of individually superior but more similar solvers, because selection correlates errors. Tiered direct democracy takes a deliberately hybrid position in this debate. Against Brennan, it retains the universal basic franchise and treats status as task-scoped rather than person-scoped. Against pure sortition, it selects on demonstrated domain knowledge rather than chance. Against flat direct democracy and the Swiss threshold regime, it grades agenda friction by status. And against all precedents, it treats competence as an investment good: status must be earned, decays, and can be re-earned, so the institution prices civic learning rather than assuming a static distribution of it.

4 The Institutional Design

4.1 Franchise architecture

The constitution partitions the binding agenda into three domain classes, each paired with an electorate. Domain A (basic civic questions: rights, values, ordinary legislation, constitutional amendments) is decided by the basic franchise, which comprises every citizen on the voter roll and is unconditional. Domain B (intermediate technical questions: budgets, infrastructure, planning) is decided by holders of tier-2 status. Domain C (highly technical questions: monetary policy parameters, nuclear safety, pharmaceutical approval) is decided by holders of tier-3 status. Figure 1 summarizes the architecture. Two invariants are constitutive rather than parametric: the basic franchise may not be removed from any citizen for failure to certify, and domain A may not be reassigned to a restricted electorate. These invariants answer, in institutional form, the objection that competence screening necessarily produces graded personhood.



Agenda access: any citizen may initiate at their own tier; thresholds fall with initiator status (x/y, x/z).

Non-zero-sum status: criterion-referenced cutoffs let the certified share grow as citizens invest.

Figure 1: The three-tier franchise architecture of tiered direct democracy, with tiered referendum thresholds and the democratic veto overlay.

Parameter	Default	Range	Notes
Basic franchise	All citizens	Fixed	Unconditional; one person, one vote on domain A
Tier-2 pool share q_2	30%	20-40%	Criterion-referenced cutoff, not a fixed quota
Tier-3 pool share q_3	10%	5-15%	Advanced certification in a technical domain
Test validity target v	0.7	At least 0.5	Squared correlation with domain competence
Status sunset	6 years	4-10 years	Re-certification required; status lapses otherwise
Domain scope	Per domain	Per domain	Energy certification confers no health status
Test governance	Open item bank	Mandatory	Adversarial review, random audits, impact monitoring

Table 1: Formal specification of the franchise and certification system

4.2 Status acquisition and decay

Status is acquired by examination, and every property of the examination is chosen to resist the historical failure modes of franchise tests. Tests are open: any citizen may attempt one in each semi-annual window, at no fee, with published curricula and a public item bank from which examination items are sampled. Cutoffs are criterion-referenced rather than norm-referenced: the passing standard is fixed in advance against the domain's competence requirements, so the certified share of the population can grow without bound as citizens respond to the incentive. This choice has a consequence the simulation in Section 5.6 confirms: status expansion is not zero-sum against the uncertified. Certifications are domain-scoped, so a citizen certified in energy economics holds no voting status on epidemiological questions. All status decays: certification lapses after six years absent re-examination, insulating the system against the accumulation of a

permanent certified class. Test governance is itself institutionalized: item banks are published, drafts undergo adversarial review by parties and civil-society organizations, administrations are randomly audited, and cutoffs are monitored for disparate impact, with results published annually. Section 6 argues that these governance mechanisms, not the arithmetic of the tiers, are the load-bearing walls of the design.

4.3 Tiered referendums and initiatives

The initiative process mirrors the franchise architecture and is pooled by initiator. Any citizen may launch a basic referendum by collecting x signatures from the citizenry at large within a fixed window T ; status-holders who sign a basic petition count as ordinary signatories, one signature each, with no added weight. A tier-2 member may launch a tier-2 referendum requiring x/y signatures from tier-2 members within the same window T ; a tier-3 member may launch a tier-3 referendum requiring x/z signatures from tier-3 members. The divisors y and z are constitutional parameters, and the simulation results in Section 5.4 recommend z greater than y , because geometrically smaller pools collect signatures more slowly and an equal divisor would starve the tier-3 agenda of access entirely. The economic logic of the discount is deliberate: lower agenda friction is the recurring rent earned by certification, and it is the mechanism that converts the design from a static filter into an incentive system.

Initiator	Eligible signers	Threshold	Window	Simulated pass rate	Median days
Any citizen	All citizens	$x = 5\%$ of N	$T = 180$ d	100%	61
Tier-2 member	Tier-2 members	x/y ($y = 2$)	$T = 180$ d	100%	89
Tier-3 member	Tier-3 members	x/z ($z = 5$)	$T = 180$ d	99.9%	99

Table 2: Referendum and initiative rules (simulated defaults)

4.4 Safeguards

Three safeguards bound the restricted electorates. First, a **democratic veto overlay**: any domain B or domain C decision may be overturned by the general electorate, by simple referendum, within one electoral cycle, whenever a petition gathering the basic threshold x demands it and a 55% supermajority of the general franchise votes to overturn. Popular sovereignty is thus preserved in the last instance; restricted electorates hold agenda power, not final authority. Second, **publication duties**: all tier ballots, agendas, and minority reports are published before decisions take effect, supplying the informational raw material on which a veto campaign depends. Third, **audit and recall**: the same random audit machinery that monitors the tests monitors the electorates, and status is revocable for demonstrated corruption. The veto overlay deserves emphasis because the simulation in Section 5.5 shows it is not merely decorative: it changes the capture calculus of the restricted paths from nontrivial to negligible, at the cost of one acknowledged failure mode, namely that a general electorate under strong misinformation may fail to wield it.

5 Simulation Model and Results

5.1 Model

We evaluate the design with a stylized agent-based simulation. A population of $N = 2,000$ citizens is drawn with a latent competence θ from a standard normal, an engagement propensity correlated with competence at 0.3, and a certification test score equal to a validity-weighted mixture of competence and noise, where the validity v is the squared correlation between score and competence. Issues arrive in three domains with mean individual accuracy targets of 0.60, 0.54, and 0.47 for domains A, B, and C respectively; the domain C target deliberately places the median citizen below chance, reflecting the systematic-deficit evidence reviewed in Section 2. Errors are correlated through shared per-issue shocks (misinformation waves) whose strength rises with domain technicality, and selected electorates suffer an additional within-group error correlation that grows with selection stringency, implementing the Hong-Page penalty in stylized form. Majority rule decides each issue; turnout follows engagement. Four systems are compared on paired draws: pure direct democracy (DD), the two-tier design (T2), the three-tier design (T3), and a standing citizens' assembly of 500 members (SORT) that receives a briefing uplift of 0.40 standard deviations in competence, following the deliberative polling evidence. Table 3 lists parameters. All results derive from 240-300 replications per configuration under a fixed seed; scripts are published with the paper. The model is a mechanism-level abstraction, not a forecast of any polity.

Parameter	Value	Meaning
Population N	2,000	Citizens per replication
Agenda mix (A, B, C)	45 / 35 / 20%	Share of issues per domain
Individual accuracy (A, B, C)	0.60 / 0.54 / 0.47	Mean citizen correctness by domain
Correlated error (A, B, C)	0.15 / 0.40 / 0.50	Shared misinformation shock by domain
Engagement	mean 0.55	Correlated 0.3 with competence
Tier cutoffs (q_2 , q_3)	30% / 10%	Criterion shares of test-score distribution
Groupthink beta	0.30	Selection penalty scaling (swept to 1.0)
Assembly briefing uplift	+0.40 SD	SORT benchmark (Fishkin-style)

Table 3: Simulation parameters (defaults)

5.2 Decision quality by domain

Figure 2 reports decision accuracy by domain at test validity 0.7. Direct democracy is essentially infallible on domain A (0.999), confirming that mass electorates are excellent judges of basic civic questions when individual accuracy is comfortably above one half, and its failure profile is exactly as the theory of Section 2 predicts: accuracy degrades to 0.765 on intermediate questions and to 0.536 on highly technical questions, indistinguishable from a coin flip weighted against the correct answer once correlated error is present. The two-tier design restores technical accuracy to 0.954; the three-tier design reaches 0.992. The briefed assembly, despite its deliberative uplift, reaches only 0.715 on domain C, suggesting that a fixed briefing cannot substitute for deep domain certification. Table 4 summarizes aggregate welfare, weighting domains by their agenda shares. The comparison is paired: every system votes on identical populations, issues, and shock realizations, so differences are attributable to the franchise structure alone.

0.536 to 0.992

Decision accuracy on highly technical questions: direct democracy versus the three-tier design (test validity $v = 0.7$)

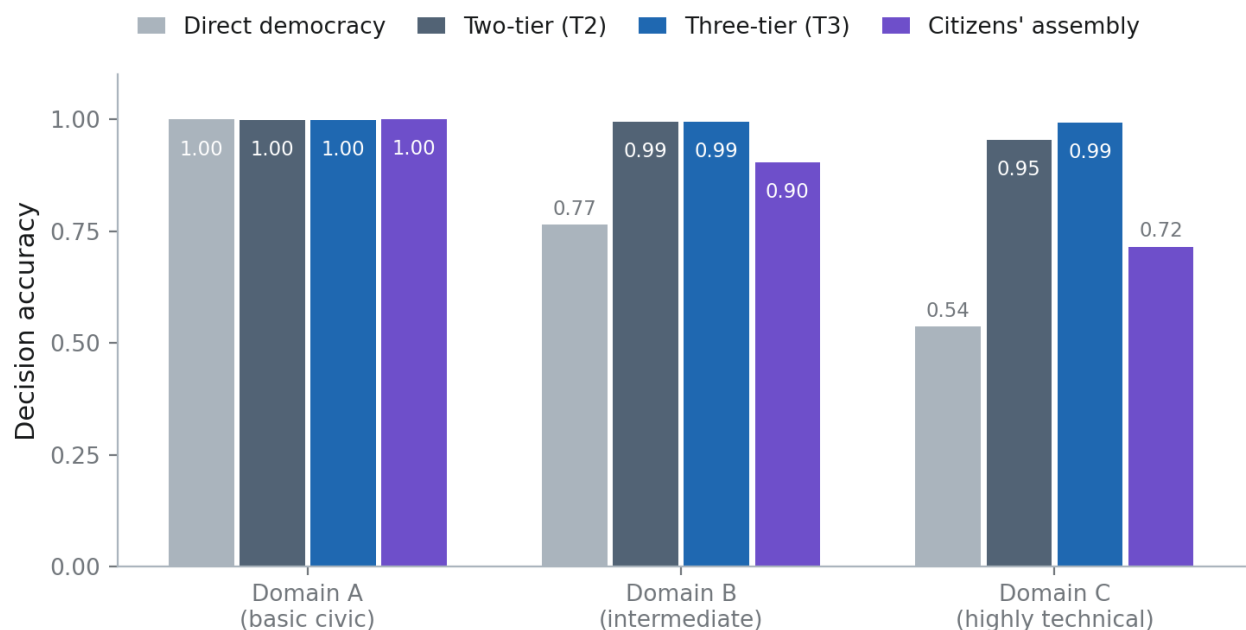


Figure 2: Decision accuracy by domain and system at test validity $v = 0.7$. Direct democracy is near-perfect on basic civic questions and near chance on highly technical ones; tiered designs recover the technical gap.

System	Domain A	Domain B	Domain C	Welfare	Voters per C-issue
Direct democracy (DD)	0.999	0.765	0.536	0.825	1,089
Two-tier (T2)	0.999	0.995	0.954	0.989	375
Three-tier (T3)	0.999	0.994	0.992	0.996	133
Citizens' assembly (SORT)	1.000	0.904	0.715	0.909	460

Table 4: Headline results at $v = 0.7$ (accuracy by domain; welfare is the agenda-weighted average)

5.3 Certification validity and the price of asymmetry

Figure 3 sweeps test validity from 0.1 to 0.9. The three-tier design dominates the two-tier design at every validity level, and the welfare gap is largest where validity is lowest (+1.5 percentage points at $v = 0.3$, against +0.6 at $v = 0.9$). The mechanism is straightforward: a more stringent cutoff extracts more expected competence per unit of test signal, so the third tier insures against certification noise. This bears directly on the practical objection that tests will be imperfect: within the model, the asymmetric design is most valuable precisely when they are. Domain C accuracy under the three-tier design exceeds 0.94 even at $v = 0.3$, whereas the two-tier design achieves 0.87 at the same validity. Direct democracy and the assembly benchmark are flat in v , as they involve no test.

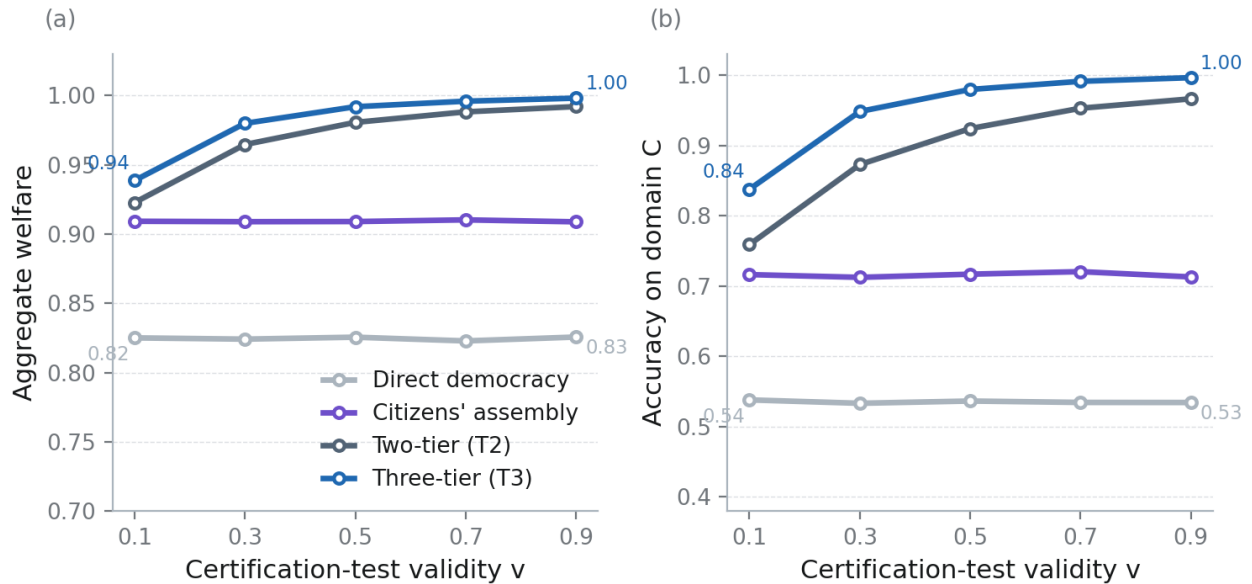


Figure 3: Aggregate welfare (a) and domain C accuracy (b) as functions of certification-test validity. The three-tier advantage over the two-tier design is largest when tests are noisy.

Figure 4 examines the selection penalty directly. At $v = 0.7$, accuracy on domain C rises monotonically as the tier-3 pool shrinks from 30% to 2% of the population, and the advantage survives groupthink coefficients up to 1.0, roughly tripling the baseline penalty, at which point a 10% pool still achieves 0.912. We also steel-manned the two-tier design by sweeping its single status cutoff. A two-tier system whose one pool is

drawn at the 10% cutoff nearly ties the three-tier design on domain C (0.992 against 0.993) and slightly exceeds it on domain B. The price of that configuration is paid elsewhere: it assigns a tenth of the citizenry to decide 55% of the agenda, doubling the legitimacy asymmetry, and the capture simulations of Section 5.5 show its bribery surface is markedly larger. The asymmetry of the three-tier design is therefore best understood not as a raw accuracy device but as a way of matching franchise breadth to epistemic demand domain by domain, buying back accuracy where it is cheap and breadth where it is valuable.

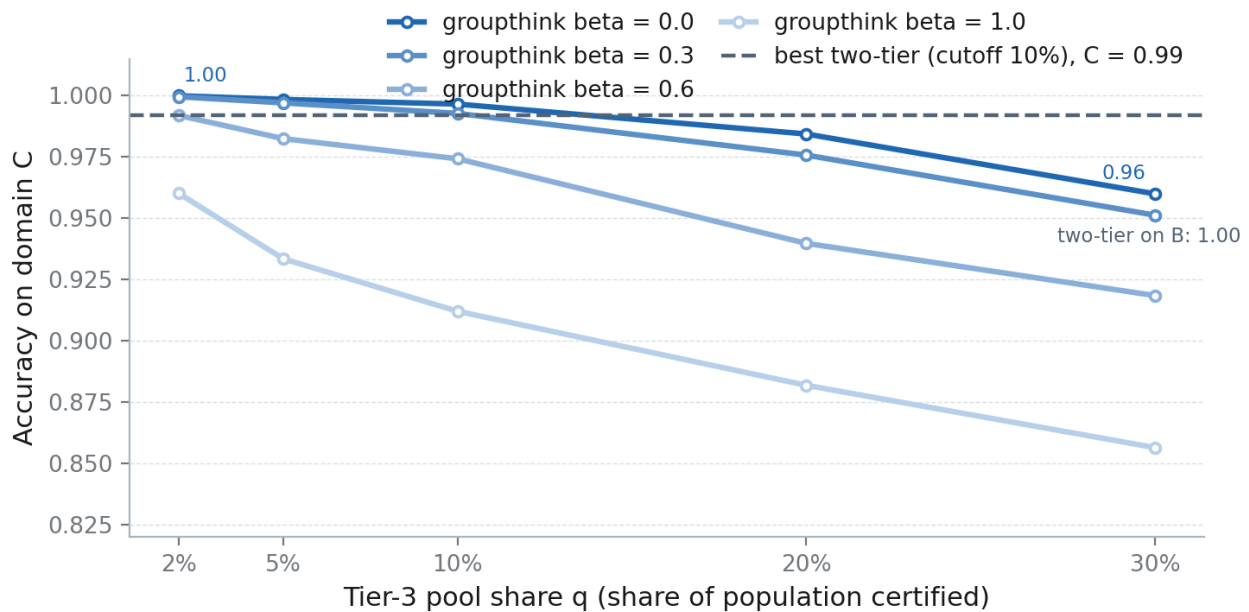


Figure 4: Domain C accuracy against the tier-3 pool share, for groupthink coefficients from 0 to 1.0 ($v = 0.7$). The dashed reference marks the steel-manned two-tier design at a 10% cutoff.

5.4 Referendum agenda access

Figure 5 reports the signature-collection experiment, in which initiatives are Poisson arrival processes whose rates depend on pool size, support, and the pool's engagement. Under the raw rule $y = z = 2$, the tier-3 path is effectively closed: the median tier-3 initiative fails to reach its 50-signature threshold inside the 180-day window, with a pass rate below one percent, because a pool of 200 members signs geometrically more slowly than the general population regardless of the divisor. Raising the tier-3 divisor to $z = 5$ restores full agenda access (99.9% pass rate, median 99 days) while keeping tier-3 initiatives slower and more deliberate than basic ones. The design lesson is that the divisors must track pool geometry rather than remaining uniform, and the defaults in Table 2 reflect it. Agenda access remains asymmetric by construction, which is the point of the incentive, but the asymmetry is bounded: no path is closed, and the basic path remains the fastest for any citizen.

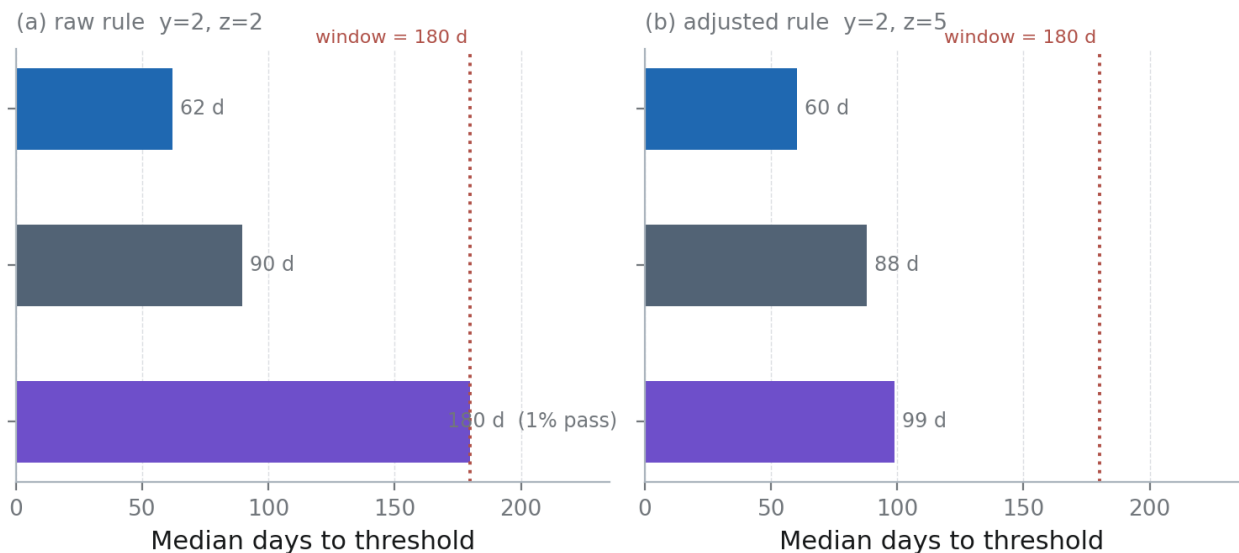


Figure 5: Median days to signature threshold by referendum path. Under the raw rule (a) the tier-3 path starves (0.8% pass rate); the adjusted divisor $z = 5$ (b) restores access within the window.

5.5 Capture resistance

Figure 6 reports the probability that a welfare-reducing measure passes, for six attack profiles, with and without the veto overlay. The asymmetry across attack channels is stark. Mass persuasion, the cheapest weapon money can buy against a general electorate (Gilens & Page, 2014), succeeds against the basic path in 56% of ballots under weak misinformation and 73% under strong misinformation. The same attacks routed at the certified tiers almost never succeed (at most 1.1%), because misinformation penalties scale with susceptibility and the certified are the least susceptible. The tiers' genuine vulnerability is bribery of a small pool: buying 15% of the tier-2 pool passes 20% of the time, and 15% of tier-3 passes 5.4%, confirming that stringency trades decision quality against capture surface. The veto overlay removes most of this vulnerability, reducing every tier-path capture probability below 0.5%, with one instructive exception: under strong misinformation the basic path retains a 46% capture probability even with the veto, because a misinformed general electorate is a coin flip away from failing to defend itself. The design implication is that the veto is a necessary but not sufficient safeguard; redundant defenses, and the tiers' resistance to mass persuasion, do the primary work.

0.73 to 0.011

Probability that a welfare-reducing measure passes under strong misinformation: basic referendum path versus the tier-3 path (no veto overlay)

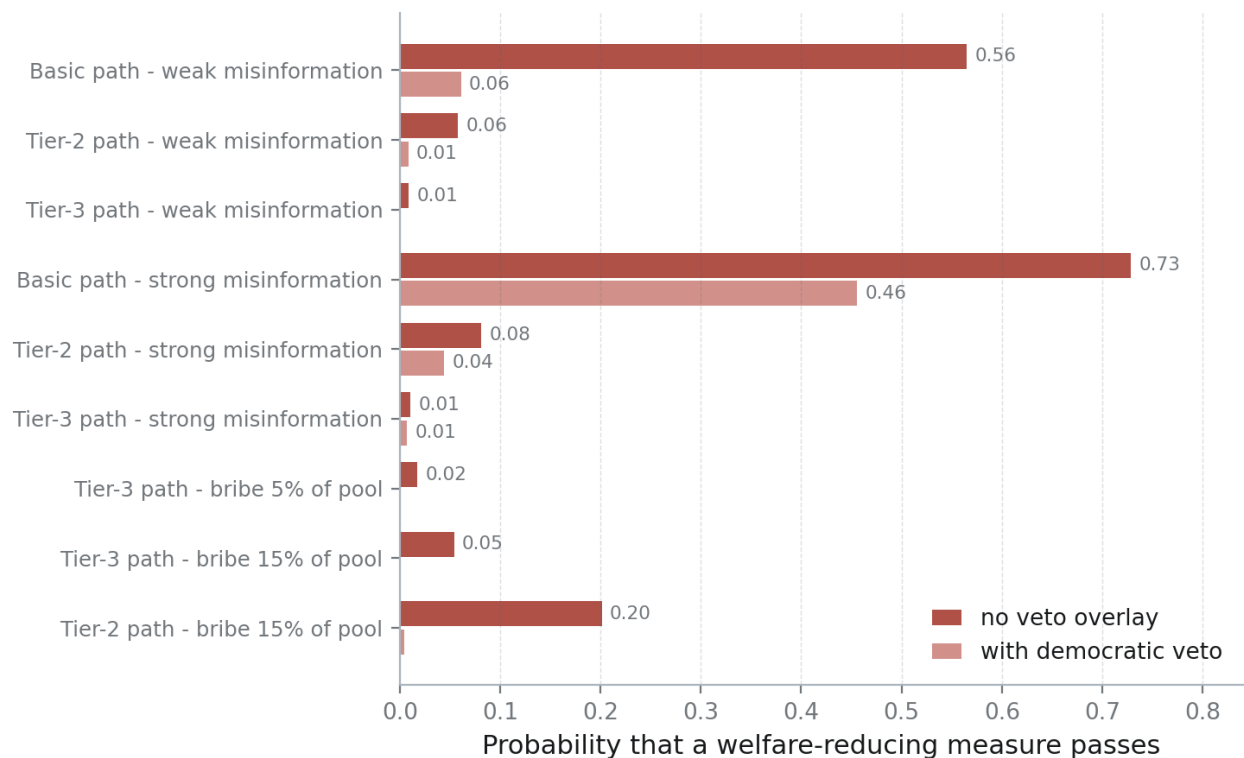


Figure 6: Capture probability by attack profile, without and with the democratic veto overlay ($v = 0.7$). Tiered paths resist mass persuasion but expose a bribery surface; the veto collapses nearly all tier-path capture below 0.5%.

5.6 Status incentives and competence growth

Figure 7 reports the incentive dynamics. Citizens choose study effort each period against the value of certification, with learning accumulating in effective competence up to a cap, and cutoffs held criterion-referenced so that the certified share may grow. Over twelve periods, the tier-2 status share expands from 31% to 47% of the population and the tier-3 share from 11% to 21%, while mean citizen competence on domain C questions rises from 0.47 to 0.61. The direct-democracy baseline, which prices civic investment at zero, is static. This is the non-zero-sum property of criterion-referenced status made visible: the rent earned by certification is paid in agenda access, and its pursuit enlarges the competent fraction of the polity rather than merely redistributing a fixed stock of status. The result should be read with the model's simplifications in mind, notably the absence of crowding costs of study and of any countervailing incentive to pursue test-taking skill rather than genuine competence.

31% to 47%

Growth of the tier-2 status share over twelve periods of status incentives; mean technical competence rises from 0.47 to 0.61 while the direct-democracy baseline stays flat

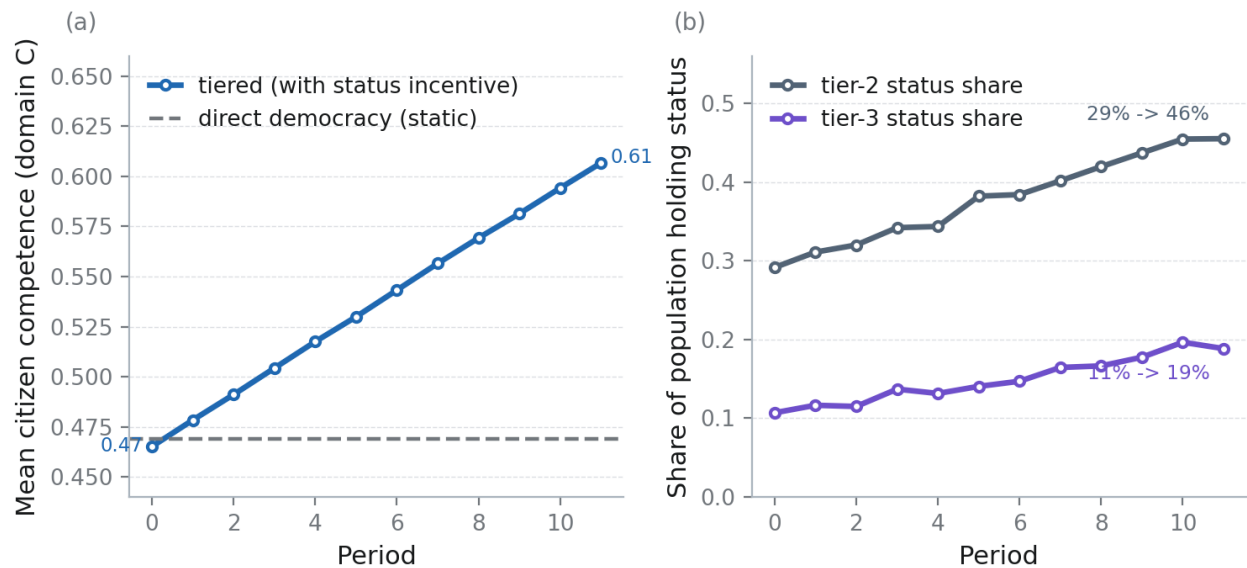


Figure 7: Incentive dynamics. (a) Mean citizen competence on domain C under tiered incentives versus the static direct-democracy baseline. (b) Status pool shares over time under criterion-referenced cutoffs.

5.7 The two-tier versus three-tier verdict

The pre-registered decision rule prefers the three-tier design only if it is not dominated by the two-tier design across the plausible validity range. The rule is satisfied: the three-tier design holds a positive welfare margin at every validity from 0.3 to 0.9, its margin on domain C accuracy is never below three percentage points, and its advantage grows as test quality falls. The steel-manned two-tier comparison of Section 5.3 narrows the raw-accuracy case to near parity, but the two-tier design then concentrates 55% of the agenda in a 10% electorate, nearly doubles the bribery capture surface, and abandons the principle that intermediate questions deserve intermediate friction. We therefore adopt the three-tier specification for the remainder of the paper, while recording that the two-tier design retains most of the accuracy gains and remains the reasonable fallback where institutional simplicity is at a premium.

6 Objections and Replies

6.1 The literacy-test specter

The gravest historical objection is that competence tests have been tried and were turned into instruments of racial exclusion. The literacy tests administered in the American South between the 1890s and the 1960s were designed, administered, and graded with discretion precisely so that Black applicants failed and white applicants passed, a history documented in detail by Keyssar (2000). Any proposal that reintroduces testing into the franchise must explain why its tests would not travel the same road. The reply is institutional rather than mathematical. The design's tests are criterion-referenced with published passing standards, drawn from a public item bank, subject to adversarial review by interested parties before adoption, randomly audited in

administration, and monitored for disparate impact with mandatory annual publication. Each of these mechanisms removes one degree of discretion that historical tests left open. Two further structural differences matter: certification here gates only technical domains, never the basic franchise, so failure excludes no one from personhood-level participation; and the simulation results quantify the cost of degradation, showing that low-validity testing forfeits most accuracy gains. We record one concession plainly: if the governance machinery cannot be sustained, the design should not be adopted, and the literacy-test record is the right warning about what failure looks like.

6.2 Political equality

A second objection holds that graded franchises violate the equal-status principle of democracy: citizens are entitled to participate as equals, not as holders of differentially valuable votes (Dahl, 1989). The reply distinguishes person-scoped from task-scoped inequality. Brennan's plural voting grades persons; this design assigns electorates to question classes, in the way that existing democracies already assign juries, medical boards, and judicial panels to their proper business without ceasing to regard citizens as political equals. Every citizen retains the unconditional basic franchise, any citizen may certify into any tier, the veto overlay preserves final authority in the general electorate, and status decays so that no permanent caste forms. The simulation adds an egalitarian datum in an unexpected place: the incentive analysis shows the certified share growing toward half the population, which over time narrows rather than entrenches the boundary. A residual expressive inequality remains, and we do not claim it away: some citizens will experience domain-restricted agendas as a standing demotion even under full formal equality of opportunity to certify.

6.3 The boundary problem

If questions are sorted into domains, the sorting power is itself sovereign, and an adversary who controls the classification can bury unwelcome questions in a restricted domain or launder favored ones into the general agenda. This is, in our judgment, the design's most serious unresolved exposure, and the honest reply is only partially reassuring. The proposed mitigation stack has four layers. Classification is fixed by published statute, not administrative discretion. A conservative default rule sends genuinely disputed classifications to the less restrictive domain, pricing capture attempts at the cost of a wider franchise. Classifications are reviewable by ordinary basic referendum on demand at the basic threshold x . And classifications sunset, forcing periodic re-justification before the general electorate. The simulation does not model classification capture directly, which we flag in Section 8; the veto overlay blunts but does not eliminate it, since a captured classification may never generate a decision that triggers a veto in the first place.

6.4 Diversity loss

Hong and Page (2004) prove that groups of diverse problem solvers can outperform groups of individually abler ones, because selection on ability correlates errors and shrinks the solution space explored. Applied here, the objection says certified electorates may be less collectively intelligent than their individual scores suggest. The design's first reply is architectural: domain scoping prevents the selection machinery from correlating errors across unrelated domains, and the broad tier-2 pool of roughly 30% preserves most of the

population's cognitive diversity on intermediate questions. The second reply is empirical: the groupthink sweep in Figure 4 shows the three-tier advantage survives error-correlation penalties three times the calibrated baseline. The third reply is a concession: the simulation models competence as a single scalar, and therefore cannot represent the functional diversity, perspective heterogeneity, and problem-framing variety on which the Hong-Page result actually operates. Our accuracy figures should accordingly be read as an upper bound on certified-electorate performance, and the diversity objection remains the strongest theory-grounded reason the tier-3 pool should not be pushed below roughly a tenth of the population.

6.5 Legitimacy asymmetry and resentment

A fifth objection concerns compliance: citizens bound by technical decisions they were excluded from voting on may withhold legitimacy, escalate conflict, and convert an epistemic institution into a symbolic grievance, in the way that independent central banks periodically become populist targets. The design replies with the veto overlay, which converts every restricted-electorate decision into a provisional ruling the general electorate may overturn, and with the sunset rule, which forces each restriction to earn its renewal. The incentive gradient blunts the boundary further: the boundary is not a wall but a queue, and the simulation shows the queue moving. What the design cannot answer is the case where certification access is itself unequal in practice, for example through unequal education or time; the no-fee, published-curriculum, semi-annual attempt structure is designed against precisely this, and Section 8 returns to it.

6.6 Elite capture

The final objection is that certified tiers will be captured by the wealthy, whether by purchasing certification preparation, corrupting test governance, or buying votes outright. The simulation prices the last of these directly: bribery of 15% of a 30% pool succeeds in about a fifth of attempts without the veto, and in under one percent with it, while mass persuasion, the historically dominant capture channel, is nearly inert against certified electorates. The structural reply to the remaining channels is pool size and churn: a 30% pool with semi-annual open examinations and six-year sunset is an expensive object to purchase or colonize, and the same audit machinery that polices disparate impact polices organized abuse of the item bank. We note that capture of the test itself collapses into the first objection's territory, which is why both replies converge on the same institutional claim: the arithmetic of tiers is the easy part, and the test-governance constitution is the part on which the design lives or dies.

7 Comparison with Alternative Systems

Table 5 locates tiered direct democracy against the main alternatives on six criteria: the legitimacy of basic-rights decisions, technical competence, participation breadth, agenda access, capture resistance, and the incentive structure each system creates for civic investment. The scores are qualitative syntheses of the simulation results where available and of the literature otherwise, and they should be read as a design-space map rather than a ranking. Three observations follow from the table. First, no competitor prices civic investment at all: every existing system treats competence as an endowment, while the tiered designs convert it into a produced good, which the dynamics experiment suggests is not a marginal feature. Second, the three-tier design's profile is dominated only on participation breadth and agenda symmetry, which are exactly the dimensions the veto overlay and the growing status share are designed to buffer. Third, the epistocracy row clarifies what the present design is not: it concedes nothing on the basic franchise, and its restricted electorates hold provisional rather than final authority, which is the institutional difference between task licensing and graded citizenship.

System	Basic rights legitimacy	Technical competence	Participation breadth	Agenda access	Capture resistance	Invests in civic competence
Representative democracy	High	Medium	Low, episodic	Low, party-gated	Medium	No
Direct democracy (DD)	High	Low	High	High, uniform	Low	No
Liquid democracy	High	Medium	High, opt-in	High	Medium	Weakly
Epistocracy (plural voting)	Contested	High	Graded	Low	Low, historically	Weakly
Sortition / open democracy	High	Medium-High	Medium, sampled	Medium	Medium	No
Two-tier TDD (T2)	High	High	Medium	Medium-High, tiered	Medium-High	Yes
Three-tier TDD (T3)	High	Highest	Medium	Medium-High, tiered	High	Yes

Table 5: Comparative assessment of democratic architectures

8 Limitations and Future Work

The simulation is deliberately stylized, and four of its simplifications bound the strength of the conclusions. First, competence is modeled as a one-dimensional scalar, so the Hong-Page diversity mechanisms discussed in Section 6.4 are represented only through an error-correlation penalty; the accuracy figures for certified electorates are therefore an upper bound. Second, test imperfection is modeled as unbiased noise, which is the favorable case: a biased instrument correlated with wealth or ethnicity would both select less competent electorates and skew representation, and no amount of tier arithmetic repairs it. Third, the boundary problem is unmodeled; classification capture must be studied with its own adversary model, and the conservative default rule proposed in Section 6.3 is an untested conjecture. Fourth, the incentive analysis omits crowding costs, strategic test-preparation that teaches to the test rather than the domain, and general-equilibrium effects on the supply of technical questions themselves. Beyond the model, two practical preconditions deserve emphasis: the design presupposes a digital identity and signature infrastructure whose integrity is a solved but nontrivial dependency, and a transition path that begins, plausibly, with a single domain C pilot, such as pharmaceutical approval or nuclear regulation, before any general reclassification of the agenda. Each of these is a concrete research program rather than a disclaimer.

9 Conclusion

This paper began with a single observation: democracies price all questions alike, and the resulting uniform epistemic friction is most costly exactly where questions are hardest. The design developed here makes the friction adaptive, retaining universal authority over basic civic questions, licensing restricted electorates for technical ones, pricing agenda access by earned status, and returning final authority to the general electorate through a veto overlay. The simulation evidence, within the acknowledged limits of a stylized model, supports three specific claims: the tiered structure recovers technical decision quality that uniform direct democracy forfeits; the finer three-way partition is preferable to the two-way one, and is most valuable when certification is imperfect; and the institution's incentive structure converts competence from an endowment into a produced good, growing the certified fraction of the polity over time. Whether any of this survives contact with a real polity depends less on the arithmetic than on whether test governance can be made honest at scale. That is an institutional question, it is answerable only by attempt, and the design's safeguards are addressed to precisely it.

References

- Achen, C. H., & Bartels, L. M. (2016). *Democracy for realists: Why elections do not produce responsive government*. Princeton University Press.
- Blum, C., & Zuber, C. I. (2016). Liquid democracy: Potentials, problems, and perspectives. *Journal of Political Philosophy*, 24(2), 162-182.
- Brennan, J. (2016). *Against democracy*. Princeton University Press.
- Caplan, B. (2007). *The myth of the rational voter: Why democracies choose bad policies*. Princeton University Press.
- Condorcet, N. C. (1785). *Essai sur l'application de l'analyse a la probabilité des décisions rendues a la pluralité des voix*. Imprimerie Royale.
- Dahl, R. A. (1989). *Democracy and its critics*. Yale University Press.
- Downs, A. (1957). *An economic theory of democracy*. Harper & Row.
- Fishkin, J. S. (2009). *When the people speak: The deliberative deficit in American democracy*. Oxford University Press.
- Gilens, M., & Page, B. I. (2014). Testing theories of American politics: Elites, interest groups, and average citizens. *Perspectives on Politics*, 12(3), 564-581.
- Hansen, M. H. (1991). *The Athenian democracy in the age of Demosthenes: Structure, principles, and ideology* (2nd ed.). University of Oklahoma Press.
- Hong, L., & Page, S. E. (2004). Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46), 16385-16389.
- Keyssar, A. (2000). *The right to vote: The contested history of democracy in the United States*. Basic Books.
- Landemore, H. (2013). *Democratic reason: Politics, collective intelligence, and the rule of the many*. Princeton University Press.
- Landemore, H. (2020). *Open democracy: Reinventing popular rule for the twenty-first century*. Princeton University Press.
- Mill, J. S. (1861). *Considerations on representative government*. Parker, Son, and Bourn.
- Ober, J. (2008). *Democracy and knowledge: Innovation and learning in classical Athens*. Princeton University Press.
- OECD. (2020). *Innovative citizen participation and new democratic institutions: Catching the deliberative wave*. OECD Publishing.
- Surowiecki, J. (2004). *The wisdom of crowds*. Doubleday.